

A. AdaBoost

Let $\text{corr}(\mathbf{x}, \mathbf{x}')$ denote the inner product (or unnormalized correlation) of two vectors $\mathbf{x}$ and $\mathbf{x}'$.

- (1) Prove that the distribution vector $(D_{t+1}(1), \dots, D_{t+1}(m))$ defined by AdaBoost and the vector of components $y_i h_t(x_i)$ are uncorrelated.
- (2) Fix $\epsilon \in (0, 1/2)$. Let the training sample be defined by m points in the plane with $\frac{m}{4}$ negative points all at coordinate $(1, 1)$, another set of $\frac{m}{4}$ negative points all at coordinate $(-1, -1)$, $\frac{m(1-\epsilon)}{4}$ positive points all at coordinate $(1, -1)$, and $\frac{m(1+\epsilon)}{4}$ positive points all at coordinate $(-1, +1)$. Describe the behavior of AdaBoost when run on this sample using boosting stumps, in particular, give the solution the algorithm returns after T rounds.

(1): Claim: The distribution vector $(D_{t+1}(1), \dots, D_{t+1}(m))$ defined by AdaBoost and the vector of components $y_i h_t(x_i)$ are uncorrelated.

Proof. Recall the AdaBoost update rule for the distribution:

$$D_{t+1}(i) = \frac{D_t(i) \exp(-\alpha_t y_i h_t(x_i))}{Z_t}$$

where Z_t is the normalization factor and $\alpha_t = \frac{1}{2} \ln \left(\frac{1-\epsilon_t}{\epsilon_t} \right)$ with $\epsilon_t = \sum_{i: y_i \neq h_t(x_i)} D_t(i)$.

To show that D_{t+1} and the vector $(y_1 h_t(x_1), \dots, y_m h_t(x_m))$ are uncorrelated, we need to prove:

$$\text{corr}(D_{t+1}, y \cdot h_t) = \sum_{i=1}^m D_{t+1}(i) \cdot y_i h_t(x_i) = 0$$

Computing this sum:

$$\sum_{i=1}^m D_{t+1}(i) \cdot y_i h_t(x_i) = \sum_{i=1}^m \frac{D_t(i) \exp(-\alpha_t y_i h_t(x_i))}{Z_t} \cdot y_i h_t(x_i)$$

Since $y_i h_t(x_i) \in \{-1, +1\}$, we can split into two cases:

Case 1: When $y_i h_t(x_i) = +1$ (correct classification):

$$\text{Contribution} = \frac{D_t(i) e^{-\alpha_t}}{Z_t} \cdot (+1)$$

Case 2: When $y_i h_t(x_i) = -1$ (incorrect classification):

$$\text{Contribution} = \frac{D_t(i) e^{\alpha_t}}{Z_t} \cdot (-1)$$

Therefore:

$$\begin{aligned}\sum_{i=1}^m D_{t+1}(i) \cdot y_i h_t(x_i) &= \frac{e^{-\alpha_t}}{Z_t} \sum_{i: y_i h_t(x_i)=1} D_t(i) - \frac{e^{\alpha_t}}{Z_t} \sum_{i: y_i h_t(x_i)=-1} D_t(i) \\ &= \frac{e^{-\alpha_t}}{Z_t} (1 - \epsilon_t) - \frac{e^{\alpha_t}}{Z_t} \epsilon_t\end{aligned}$$

Now, using $\alpha_t = \frac{1}{2} \ln \left(\frac{1-\epsilon_t}{\epsilon_t} \right)$:

$$e^{\alpha_t} = \sqrt{\frac{1-\epsilon_t}{\epsilon_t}}, \quad e^{-\alpha_t} = \sqrt{\frac{\epsilon_t}{1-\epsilon_t}}$$

Substituting:

$$\begin{aligned}&= \frac{1}{Z_t} \left[\sqrt{\frac{\epsilon_t}{1-\epsilon_t}} (1 - \epsilon_t) - \sqrt{\frac{1-\epsilon_t}{\epsilon_t}} \epsilon_t \right] \\ &= \frac{1}{Z_t} \left[\sqrt{\epsilon_t(1-\epsilon_t)} - \sqrt{\epsilon_t(1-\epsilon_t)} \right] \\ &= 0\end{aligned}$$

Thus, $\text{corr}(D_{t+1}, y \cdot h_t) = 0$, proving that the vectors are uncorrelated. $\square$

(2):

Dataset: m points with $\frac{m}{4}$ negative at $(1, 1)$, $\frac{m}{4}$ negative at $(-1, -1)$, $\frac{m(1-\epsilon)}{4}$ positive at $(1, -1)$, and $\frac{m(1+\epsilon)}{4}$ positive at $(-1, 1)$.

Relevant decision stumps:

- $h^{(1)}(x) = \text{sign}(x_2)$: predicts +1 if $x_2 > 0$
- $h^{(2)}(x) = -\text{sign}(x_1)$: predicts +1 if $x_1 < 0$

Round 1: With uniform distribution $D_1(i) = \frac{1}{m}$, we compute weighted errors:

$$\begin{aligned}\epsilon(h^{(1)}) &= \frac{1}{m} \left(\frac{m}{4} + \frac{m(1-\epsilon)}{4} \right) = \frac{1}{2} - \frac{\epsilon}{4} \\ \epsilon(h^{(2)}) &= \frac{1}{m} \left(\frac{m}{4} + \frac{m(1-\epsilon)}{4} \right) = \frac{1}{2} - \frac{\epsilon}{4}\end{aligned}$$

Both achieve minimum error. Suppose $h_1 = h^{(1)}$ is selected with weight:

$$\alpha_1 = \frac{1}{2} \ln \left(\frac{2+\epsilon}{2-\epsilon} \right)$$

Round 2: After weight update, misclassified points (at $(1, 1)$ and $(1, -1)$) have increased weights. The optimal stump becomes $h_2 = h^{(2)} = -\text{sign}(x_1)$.

Alternating pattern: Due to the symmetric structure, AdaBoost alternates between $h^{(1)}$ (based on x_2) in odd rounds and $h^{(2)}$ (based on x_1) in even rounds.

Solution after T rounds:

$$H_T(x) = \text{sign} \left(\sum_{k=1}^{\lceil T/2 \rceil} \alpha_{2k-1} \cdot \text{sign}(x_2) + \sum_{k=1}^{\lfloor T/2 \rfloor} \alpha_{2k} \cdot (-\text{sign}(x_1)) \right)$$

This can be written as:

$$H_T(x) = \text{sign}(\beta_1 \cdot \text{sign}(x_2) - \beta_2 \cdot \text{sign}(x_1))$$

where $\beta_1 = \sum_{t \text{ odd}} \alpha_t$ and $\beta_2 = \sum_{t \text{ even}} \alpha_t$.

The combined classifier represents a linear decision boundary passing through the origin, with slope determined by β_2/β_1 . The asymmetry parameter ϵ causes the boundary to tilt slightly to accommodate the imbalanced positive class distribution.

B. Boosting algorithms

- (1) Show that for all $u \in \mathbb{R}$ and integer $p > 1$, $1_{u \leq 0} \leq \Psi_p(-u)$ where $\Psi_p(u) = \max((1+u)^p, 0)$. Show that Ψ_p is convex and differentiable.
- (2) Use Ψ_p to derive a boosting-type algorithm using coordinate descent. You should give a full description of your algorithm, including the pseudocode, details for the choice of the step and direction, as well as a generalization bound.

(1)

Part (a): Show that $1_{u \leq 0} \leq \Psi_p(-u)$ for all $u \in \mathbb{R}$ and integer $p > 1$.

Proof. We have $\Psi_p(u) = \max((1+u)^p, 0)$, so $\Psi_p(-u) = \max((1-u)^p, 0)$.

We consider two cases:

Case 1: $u > 0$ (i.e., $u \leq 0$ is false)

Then $1_{u \leq 0} = 0$. Since $\Psi_p(-u) \geq 0$ by definition, we have $0 \leq \Psi_p(-u)$. ✓

Case 2: $u \leq 0$

Then $1_{u \leq 0} = 1$. Since $u \leq 0$, we have $-u \geq 0$, so $1-u \geq 1$. Thus:

$$(1-u)^p \geq 1^p = 1$$

Therefore $\Psi_p(-u) = (1-u)^p \geq 1 = 1_{u \leq 0}$. ✓

In both cases, $1_{u \leq 0} \leq \Psi_p(-u)$. □

Part (b): Show that Ψ_p is convex.

Proof. For $u \geq -1$: $\Psi_p(u) = (1 + u)^p$. The second derivative is:

$$\Psi_p''(u) = p(p-1)(1+u)^{p-2} \geq 0 \quad \text{for } u \geq -1, p > 1$$

For $u < -1$: $\Psi_p(u) = 0$, which is trivially convex.

At $u = -1$: The function transitions from $(1 + u)^p$ to 0. Since $(1 + u)^p \rightarrow 0$ as $u \rightarrow -1^+$ and the function equals 0 for $u < -1$, the function is continuous.

For convexity at the junction, we verify that for any $\lambda \in [0, 1]$ and $u_1 < -1 < u_2$:

$$\Psi_p(\lambda u_1 + (1 - \lambda)u_2) \leq \lambda \Psi_p(u_1) + (1 - \lambda)\Psi_p(u_2) = (1 - \lambda)(1 + u_2)^p$$

This holds because $(1 + \lambda u_1 + (1 - \lambda)u_2)^p \leq (1 - \lambda)^p(1 + u_2)^p \leq (1 - \lambda)(1 + u_2)^p$ for $p > 1$.

Thus Ψ_p is convex on $\mathbb{R}$. □

Part (c): Show that Ψ_p is differentiable.

Proof. For $u > -1$: $\Psi_p'(u) = p(1 + u)^{p-1}$.

For $u < -1$: $\Psi_p'(u) = 0$.

At $u = -1$:

$$\lim_{u \rightarrow -1^+} \Psi_p'(u) = \lim_{u \rightarrow -1^+} p(1 + u)^{p-1} = 0 = \lim_{u \rightarrow -1^-} \Psi_p'(u)$$

Since the left and right derivatives agree at $u = -1$, Ψ_p is differentiable everywhere with:

$$\Psi_p'(u) = \begin{cases} p(1 + u)^{p-1} & \text{if } u \geq -1 \\ 0 & \text{if } u < -1 \end{cases}$$

□

(2)

Objective Function

We minimize the empirical loss using Ψ_p as a surrogate for the 0-1 loss:

$$L(\alpha) = \frac{1}{m} \sum_{i=1}^m \Psi_p(-y_i f(x_i)) = \frac{1}{m} \sum_{i=1}^m \Psi_p \left(-y_i \sum_{j=1}^N \alpha_j h_j(x_i) \right)$$

where $f(x) = \sum_{j=1}^N \alpha_j h_j(x)$ is a linear combination of base hypotheses.

Coordinate Descent Approach

At each round t , we select a hypothesis h_t and step size η_t to update:

$$f_t(x) = f_{t-1}(x) + \eta_t h_t(x)$$

Choosing the Direction (Hypothesis Selection)

The gradient of L with respect to adding $\eta \cdot h_j$ is:

$$\left. \frac{\partial L}{\partial \eta} \right|_{\eta=0} = -\frac{1}{m} \sum_{i=1}^m y_i h_j(x_i) \Psi'_p(-y_i f_{t-1}(x_i))$$

Define weights:

$$w_i^{(t)} = \Psi'_p(-y_i f_{t-1}(x_i)) = p(1 - y_i f_{t-1}(x_i))^{p-1} \cdot 1_{y_i f_{t-1}(x_i) \leq 1}$$

Select the hypothesis h_t that maximizes the weighted correlation:

$$h_t = \arg \max_{h \in \mathcal{H}} \left| \sum_{i=1}^m w_i^{(t)} y_i h(x_i) \right|$$

Choosing the Step Size

Using line search, find η_t that minimizes:

$$\eta_t = \arg \min_{\eta} \frac{1}{m} \sum_{i=1}^m \Psi_p(-y_i(f_{t-1}(x_i) + \eta h_t(x_i)))$$

For a closed-form approximation, use Newton's method or a fixed learning rate $\eta_t = \eta$.

Algorithm: Ψ_p -Boost

Algorithm 1 Ψ_p -Boost

Require: Training set $S = \{(x_i, y_i)\}_{i=1}^m$, hypothesis class $\mathcal{H}$, rounds T , parameter $p > 1$

- 1: Initialize $f_0(x) = 0$
 - 2: **for** $t = 1$ to T **do**
 - 3: Compute weights: $w_i^{(t)} = p(1 - y_i f_{t-1}(x_i))^{p-1} \cdot 1_{y_i f_{t-1}(x_i) \leq 1}$
 - 4: Select hypothesis: $h_t = \arg \max_{h \in \mathcal{H}} \left| \sum_{i=1}^m w_i^{(t)} y_i h(x_i) \right|$
 - 5: Determine sign: $s_t = \text{sign} \left(\sum_{i=1}^m w_i^{(t)} y_i h_t(x_i) \right)$
 - 6: Find step size: $\eta_t = \arg \min_{\eta \geq 0} \sum_{i=1}^m \Psi_p(-y_i(f_{t-1}(x_i) + \eta \cdot s_t \cdot h_t(x_i)))$
 - 7: Update: $f_t(x) = f_{t-1}(x) + \eta_t \cdot s_t \cdot h_t(x)$
 - 8: **end for**
 - 9: **return** $H(x) = \text{sign}(f_T(x))$
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Generalization Bound

Let $\mathcal{H}$ be a hypothesis class with VC-dimension d . After T rounds, the combined hypothesis H_T belongs to the class of linear combinations of T base hypotheses.

Theorem 0.1. With probability at least $1 - \delta$ over the sample S of size m , for all margins $\theta > 0$:

$$R(H_T) \leq \hat{R}_\theta(H_T) + O \left(\sqrt{\frac{Td \log(m/d) + \log(1/\delta)}{m}} \right)$$

where $R(H_T)$ is the true error, and $\hat{R}_\theta(H_T) = \frac{1}{m} \sum_{i=1}^m 1_{y_i f_T(x_i) \leq \theta}$ is the empirical margin error.

Alternatively, using Rademacher complexity:

Theorem 0.2. Let $\mathcal{R}_m(\mathcal{H})$ denote the Rademacher complexity of $\mathcal{H}$. With probability at least $1 - \delta$:

$$R(H_T) \leq \frac{1}{m} \sum_{i=1}^m \Psi_p(-y_i f_T(x_i)) + 2T \cdot \mathcal{R}_m(\mathcal{H}) + \sqrt{\frac{\log(1/\delta)}{2m}}$$

Since $\Psi_p(-u) \geq 1_{u \leq 0}$, minimizing the Ψ_p loss provides an upper bound on the empirical 0-1 loss, and the generalization bound ensures that the true error is controlled.

C. Regularized Maxent

This problem studies L_2 -regularized Maxent. We will use the notation introduced in class and will denote by J_S the dual objective function minimized given a sample S :

$$J_S(\mathbf{w}) = \frac{\lambda}{2} \|\mathbf{w}\|_2^2 + \mathbb{E}_{x \sim S} [-\log p_{\mathbf{w}}[x]],$$

where $\lambda > 0$ is a regularization parameter. We will assume that the feature vector is bounded: $\|\Phi(x)\|_2 \leq r$ for all $x \in \mathcal{X}$, for some $r > 0$.

- (1) Use McDiarmid's inequality to prove that for any $\delta > 0$, with probability at least $1 - \delta$, the following inequality holds:

$$\left\| \mathbb{E}_{x \sim \mathcal{D}} [\Phi(x)] - \mathbb{E}_{x \sim S} [\Phi(x)] \right\|_2 \leq \sqrt{\frac{2r^2}{m}} \left(1 + \log \sqrt{\frac{1}{\delta}} \right).$$

- (2) Let $\hat{\mathbf{w}}$ be the L_2 -regularized maxent solution for a sample S and $\mathbf{w}_{\mathcal{D}}$ the solution for an infinite sample:

$$\hat{\mathbf{w}} = \operatorname{argmin}_{\mathbf{w} \in \mathbb{R}^N} J_S(\mathbf{w}) \quad \text{and} \quad \mathbf{w}_{\mathcal{D}} = \operatorname{argmin}_{\mathbf{w} \in \mathbb{R}^N} J_{\mathcal{D}}(\mathbf{w}).$$

where $J_{\mathcal{D}}(\mathbf{w}) = \frac{\lambda}{2} \|\mathbf{w}\|_2^2 + \mathbb{E}_{x \sim \mathcal{D}} [-\log p_{\mathbf{w}}[x]]$. Use the definition of $\hat{\mathbf{w}}$ and $\mathbf{w}_{\mathcal{D}}$ as minimizers (use gradients) to prove that the following inequality holds:

$$\|\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}\|_2 \leq \frac{\|\mathbb{E}_{x \sim S} [\Phi(x)] - \mathbb{E}_{x \sim \mathcal{D}} [\Phi(x)]\|_2}{\lambda}.$$

- (3) For any $\mathbf{w}$ and any distribution $\mathcal{Q}$ define $\mathcal{L}_{\mathcal{Q}}(\mathbf{w})$ by $\mathcal{L}_{\mathcal{Q}}(\mathbf{w}) = \mathbb{E}_{x \sim \mathcal{Q}} [-\log p_{\mathbf{w}}[x]]$. Show that

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) - \mathcal{L}_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) \leq (\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) \cdot \left[\mathbb{E}_{x \sim S} [\Phi(x)] - \mathbb{E}_{x \sim \mathcal{D}} [\Phi(x)] \right] + \frac{\lambda}{2} \|\mathbf{w}_{\mathcal{D}}\|_2^2 - \frac{\lambda}{2} \|\hat{\mathbf{w}}\|_2^2.$$

- (4) Use that to show that the following inequality holds for any $\mathbf{w}$:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) \leq \frac{1}{\lambda} \left\| \mathbb{E}_{x \sim S} [\Phi(x)] - \mathbb{E}_{x \sim \mathcal{D}} [\Phi(x)] \right\|_2^2 + \mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2.$$

- (5) Conclude by proving that for any $\delta > 0$, with probability at least $1 - \delta$, the following inequality holds:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) \leq \inf_{\mathbf{w} \in \mathbb{R}^N} \mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 + \frac{2r^2}{\lambda m} \left(1 + \sqrt{\log \frac{1}{\delta}} \right)^2.$$

(1)

Claim: With probability at least $1 - \delta$:

$$\|\mathbb{E}_{x \sim \mathcal{D}}[\Phi(x)] - \mathbb{E}_{x \sim S}[\Phi(x)]\|_2 \leq \sqrt{\frac{2r^2}{m}} \left(1 + \sqrt{\log \frac{1}{\delta}}\right)$$

Proof. Define $F(x_1, \dots, x_m) = \|\mathbb{E}_{x \sim \mathcal{D}}[\Phi(x)] - \frac{1}{m} \sum_{i=1}^m \Phi(x_i)\|_2$.

When we change a single sample x_k to x'_k , by triangle inequality:

$$\begin{aligned} |F(x_1, \dots, x_k, \dots, x_m) - F(x_1, \dots, x'_k, \dots, x_m)| &\leq \frac{1}{m} \|\Phi(x_k) - \Phi(x'_k)\|_2 \\ &\leq \frac{2r}{m} \end{aligned}$$

Thus F satisfies the bounded differences condition with $c_k = \frac{2r}{m}$.

By McDiarmid's inequality, with probability at least $1 - \delta$:

$$F(S) \leq \mathbb{E}[F(S)] + \sqrt{\frac{\sum_{k=1}^m c_k^2}{2} \log \frac{1}{\delta}} = \mathbb{E}[F(S)] + \sqrt{\frac{2r^2}{m} \log \frac{1}{\delta}}$$

For the expectation term, using Jensen's inequality:

$$\mathbb{E}[F(S)] \leq \sqrt{\mathbb{E} \left[\left\| \mathbb{E}_{\mathcal{D}}[\Phi] - \frac{1}{m} \sum_{i=1}^m \Phi(x_i) \right\|_2^2 \right]} = \sqrt{\frac{\text{Var}(\Phi)}{m}} \leq \sqrt{\frac{r^2}{m}} = \frac{r}{\sqrt{m}}$$

Combining:

$$F(S) \leq \frac{r}{\sqrt{m}} + r \sqrt{\frac{2 \log(1/\delta)}{m}} = \sqrt{\frac{2r^2}{m}} \left(\frac{1}{\sqrt{2}} + \sqrt{\log \frac{1}{\delta}} \right) \leq \sqrt{\frac{2r^2}{m}} \left(1 + \sqrt{\log \frac{1}{\delta}} \right)$$

□

(2)

Claim: $\|\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}\|_2 \leq \frac{1}{\lambda} \|\mathbb{E}_{x \sim S}[\Phi(x)] - \mathbb{E}_{x \sim \mathcal{D}}[\Phi(x)]\|_2$

Proof. The gradient of the negative log-likelihood is:

$$\nabla_{\mathbf{w}} (-\log p_{\mathbf{w}}[x]) = \mathbb{E}_{x' \sim p_{\mathbf{w}}}[\Phi(x')] - \Phi(x)$$

Setting the gradients of J_S and $J_{\mathcal{D}}$ to zero at their minimizers:

$$\begin{aligned} \nabla J_S(\hat{\mathbf{w}}) = 0 &\Rightarrow \lambda \hat{\mathbf{w}} + \mathbb{E}_{x' \sim p_{\hat{\mathbf{w}}}}[\Phi(x')] - \mathbb{E}_{x \sim S}[\Phi(x)] = 0 \\ \nabla J_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) = 0 &\Rightarrow \lambda \mathbf{w}_{\mathcal{D}} + \mathbb{E}_{x' \sim p_{\mathbf{w}_{\mathcal{D}}}}[\Phi(x')] - \mathbb{E}_{x \sim \mathcal{D}}[\Phi(x)] = 0 \end{aligned}$$

Subtracting:

$$\lambda(\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) + \mathbb{E}_{p_{\hat{\mathbf{w}}}}[\Phi] - \mathbb{E}_{p_{\mathbf{w}_{\mathcal{D}}}}[\Phi] = \mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]$$

Taking the dot product with $(\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}})$:

$$\lambda \|\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}\|_2^2 + (\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) \cdot (\mathbb{E}_{p_{\hat{\mathbf{w}}}}[\Phi] - \mathbb{E}_{p_{\mathbf{w}_{\mathcal{D}}}}[\Phi]) = (\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) \cdot (\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi])$$

The mapping $\mathbf{w} \mapsto \mathbb{E}_{p_{\mathbf{w}}}[\Phi]$ is the gradient of the log-partition function, which is convex. Therefore:

$$(\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) \cdot (\mathbb{E}_{p_{\hat{\mathbf{w}}}}[\Phi] - \mathbb{E}_{p_{\mathbf{w}_{\mathcal{D}}}}[\Phi]) \geq 0$$

Thus:

$$\lambda \|\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}\|_2^2 \leq (\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) \cdot (\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]) \leq \|\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}\|_2 \cdot \|\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]\|_2$$

Dividing by $\lambda \|\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}\|_2$ yields the result. $\square$

(3)

Claim:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) - \mathcal{L}_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) \leq (\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) \cdot [\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]] + \frac{\lambda}{2} \|\mathbf{w}_{\mathcal{D}}\|_2^2 - \frac{\lambda}{2} \|\hat{\mathbf{w}}\|_2^2$$

Proof. Since $\mathbf{w}_{\mathcal{D}}$ minimizes $J_{\mathcal{D}}$, we have $J_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) \leq J_{\mathcal{D}}(\hat{\mathbf{w}})$:

$$\frac{\lambda}{2} \|\mathbf{w}_{\mathcal{D}}\|_2^2 + \mathcal{L}_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) \leq \frac{\lambda}{2} \|\hat{\mathbf{w}}\|_2^2 + \mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}})$$

Rearranging:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) - \mathcal{L}_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) \geq \frac{\lambda}{2} \|\mathbf{w}_{\mathcal{D}}\|_2^2 - \frac{\lambda}{2} \|\hat{\mathbf{w}}\|_2^2$$

Similarly, since $\hat{\mathbf{w}}$ minimizes J_S : $J_S(\hat{\mathbf{w}}) \leq J_S(\mathbf{w}_{\mathcal{D}})$:

$$\frac{\lambda}{2} \|\hat{\mathbf{w}}\|_2^2 + \mathcal{L}_S(\hat{\mathbf{w}}) \leq \frac{\lambda}{2} \|\mathbf{w}_{\mathcal{D}}\|_2^2 + \mathcal{L}_S(\mathbf{w}_{\mathcal{D}})$$

Now, the key observation is:

$$\mathcal{L}_{\mathcal{D}}(\mathbf{w}) - \mathcal{L}_S(\mathbf{w}) = \mathbf{w} \cdot (\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi])$$

This follows because $-\log p_{\mathbf{w}}[x] = \log Z(\mathbf{w}) - \mathbf{w} \cdot \Phi(x)$, so:

$$\mathcal{L}_{\mathcal{D}}(\mathbf{w}) - \mathcal{L}_S(\mathbf{w}) = -\mathbf{w} \cdot \mathbb{E}_{\mathcal{D}}[\Phi] + \mathbf{w} \cdot \mathbb{E}_S[\Phi] = \mathbf{w} \cdot (\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi])$$

Therefore:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) = \mathcal{L}_S(\hat{\mathbf{w}}) + \hat{\mathbf{w}} \cdot (\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi])$$

$$\mathcal{L}_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) = \mathcal{L}_S(\mathbf{w}_{\mathcal{D}}) + \mathbf{w}_{\mathcal{D}} \cdot (\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi])$$

Thus:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) - \mathcal{L}_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) = \mathcal{L}_S(\hat{\mathbf{w}}) - \mathcal{L}_S(\mathbf{w}_{\mathcal{D}}) + (\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) \cdot (\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi])$$

From the optimality of $\hat{\mathbf{w}}$ for J_S :

$$\mathcal{L}_S(\hat{\mathbf{w}}) - \mathcal{L}_S(\mathbf{w}_{\mathcal{D}}) \leq \frac{\lambda}{2} \|\mathbf{w}_{\mathcal{D}}\|_2^2 - \frac{\lambda}{2} \|\hat{\mathbf{w}}\|_2^2$$

Combining yields the result. $\square$

(4)

Claim: For any $\mathbf{w}$:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) \leq \frac{1}{\lambda} \|\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]\|_2^2 + \mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$$

Proof. From Problem 3:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) - \mathcal{L}_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) \leq (\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) \cdot (\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]) + \frac{\lambda}{2} \|\mathbf{w}_{\mathcal{D}}\|_2^2 - \frac{\lambda}{2} \|\hat{\mathbf{w}}\|_2^2$$

Using Cauchy-Schwarz and the result from Problem 2:

$$\begin{aligned} (\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}) \cdot (\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]) &\leq \|\hat{\mathbf{w}} - \mathbf{w}_{\mathcal{D}}\|_2 \cdot \|\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]\|_2 \\ &\leq \frac{1}{\lambda} \|\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]\|_2^2 \end{aligned}$$

Since $\|\hat{\mathbf{w}}\|_2^2 \geq 0$ and $\mathbf{w}_{\mathcal{D}}$ minimizes $J_{\mathcal{D}}$, for any $\mathbf{w}$:

$$\mathcal{L}_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) + \frac{\lambda}{2} \|\mathbf{w}_{\mathcal{D}}\|_2^2 \leq \mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$$

Therefore:

$$\begin{aligned} \mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) &\leq \mathcal{L}_{\mathcal{D}}(\mathbf{w}_{\mathcal{D}}) + \frac{1}{\lambda} \|\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]\|_2^2 + \frac{\lambda}{2} \|\mathbf{w}_{\mathcal{D}}\|_2^2 \\ &\leq \frac{1}{\lambda} \|\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]\|_2^2 + \mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 \end{aligned}$$

$\square$

(5)

Claim: With probability at least $1 - \delta$:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) \leq \inf_{\mathbf{w}} \left[\mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 \right] + \frac{2r^2}{\lambda m} \left(1 + \sqrt{\log \frac{1}{\delta}} \right)^2$$

Proof. From Problem 4, for any $\mathbf{w}$:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) \leq \frac{1}{\lambda} \|\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]\|_2^2 + \mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2$$

Taking the infimum over $\mathbf{w}$:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) \leq \frac{1}{\lambda} \|\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]\|_2^2 + \inf_{\mathbf{w}} \left[\mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 \right]$$

From Problem 1, with probability at least $1 - \delta$:

$$\|\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]\|_2 \leq \sqrt{\frac{2r^2}{m}} \left(1 + \sqrt{\log \frac{1}{\delta}} \right)$$

Therefore:

$$\|\mathbb{E}_S[\Phi] - \mathbb{E}_{\mathcal{D}}[\Phi]\|_2^2 \leq \frac{2r^2}{m} \left(1 + \sqrt{\log \frac{1}{\delta}} \right)^2$$

Substituting:

$$\mathcal{L}_{\mathcal{D}}(\hat{\mathbf{w}}) \leq \inf_{\mathbf{w}} \left[\mathcal{L}_{\mathcal{D}}(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 \right] + \frac{2r^2}{\lambda m} \left(1 + \sqrt{\log \frac{1}{\delta}} \right)^2$$

□

Interpretation: This bound shows that the expected loss of the empirical minimizer $\hat{\mathbf{w}}$ on the true distribution $\mathcal{D}$ is controlled by the regularized loss of any comparator $\mathbf{w}$, plus an estimation error term of order $O(r^2/(\lambda m))$. The trade-off between approximation (controlled by $\lambda \|\mathbf{w}\|^2$) and estimation (controlled by $1/\lambda$) is evident.

D. Randomized Halving

In class, we showed that, in the realizable scenario (at least one expert is always correct), the number of mistakes made by Halving is upper bounded by $\log_2 N$. Here, we consider for the same realizable scenario a randomized version of Halving defined as follows.

As for Halving, let H_t denote the set of remaining experts at the beginning of round t , with $H_1 = H$ the full set of N experts. At each round, let r_t be the fraction of experts in H_t predicting 1. Then, the prediction $\hat{y}_t$ made by the algorithm is 1 with probability

$$p_t = \left[\frac{1}{2} \log_2 \frac{1}{1 - r_t} \right] 1_{r_t \leq \frac{3}{4}} + 1_{r_t > \frac{3}{4}},$$

0 with probability $1 - p_t$. The true label y_t is then received and H_{t+1} is derived from H_t by removing all experts who made a mistake.

- (1) Write the pseudocode of the algorithm.
- (2) Define the potential function $\Psi_t = \log_2 |H_t|$. Let $\mu_t = \mathbf{1}_{y_t \neq \hat{y}_t}$, prove that for all $t \geq 1$, $\mathbb{E}[\mu_t] \leq \frac{\Psi_t - \mathbb{E}[\Psi_{t+1}]}{2}$.
- (3) Show that the expected number of mistakes made by randomized Halving is at most $\frac{1}{2} \log_2 N$.
- (4) (Bonus question) Prove that no randomized algorithm makes fewer than $\lfloor \frac{1}{2} \log_2 N \rfloor$ mistakes, in expectation.

(1)

Algorithm 2 Randomized Halving

```

1: Input: Set of experts  $H = \{1, 2, \dots, N\}$ 
2: Initialize:  $H_1 \leftarrow H$ 
3: for  $t = 1, 2, \dots, T$  do
4:    $n_t^{(1)} \leftarrow |\{i \in H_t : \text{expert } i \text{ predicts } 1\}|$ 
5:    $r_t \leftarrow n_t^{(1)} / |H_t|$  {Fraction predicting 1}
6:   if  $r_t > \frac{3}{4}$  then
7:      $p_t \leftarrow 1$ 
8:   else
9:      $p_t \leftarrow \frac{1}{2} \log_2 \frac{1}{1-r_t}$ 
10:  end if
11:  Draw  $\hat{y}_t \sim \text{Bernoulli}(p_t)$  {Predict 1 with probability  $p_t$ }
12:  Output prediction  $\hat{y}_t$ 
13:  Receive true label  $y_t$ 
14:   $H_{t+1} \leftarrow \{i \in H_t : \text{expert } i \text{ predicted } y_t\}$  {Remove incorrect experts}
15: end for

```

2

Claim: For all $t \geq 1$, $\mathbb{E}[\mu_t] \leq \frac{\Psi_t - \mathbb{E}[\Psi_{t+1}]}{2}$, where $\mu_t = \mathbf{1}_{y_t \neq \hat{y}_t}$.

Proof. Let $|H_t| = n_t$. Since we are in the realizable setting, there exists at least one perfect expert, so $|H_{t+1}| \geq 1$ always.

When $y_t = 1$: The experts predicting 1 survive, so $|H_{t+1}| = r_t \cdot n_t$.

When $y_t = 0$: The experts predicting 0 survive, so $|H_{t+1}| = (1 - r_t) \cdot n_t$.

The change in potential is:

$$\Psi_t - \Psi_{t+1} = \log_2 n_t - \log_2 |H_{t+1}| = \begin{cases} \log_2 \frac{1}{r_t} & \text{if } y_t = 1 \\ \log_2 \frac{1}{1-r_t} & \text{if } y_t = 0 \end{cases}$$

Case 1: $r_t \leq \frac{3}{4}$

Here $p_t = \frac{1}{2} \log_2 \frac{1}{1-r_t}$. Note that since $r_t \leq \frac{3}{4}$, we have $1 - r_t \geq \frac{1}{4}$, so $\frac{1}{1-r_t} \leq 4$, giving $p_t \leq 1$.

The probability of making a mistake:

$$\mathbb{E}[\mu_t] = \mathbb{P}(\hat{y}_t = 1, y_t = 0) + \mathbb{P}(\hat{y}_t = 0, y_t = 1) = p_t \cdot \mathbf{1}_{y_t=0} + (1 - p_t) \cdot \mathbf{1}_{y_t=1}$$

Subcase 1a: $y_t = 0$

Then $\mathbb{E}[\mu_t] = p_t = \frac{1}{2} \log_2 \frac{1}{1-r_t}$ and $\Psi_t - \Psi_{t+1} = \log_2 \frac{1}{1-r_t}$.

Thus: $\mathbb{E}[\mu_t] = \frac{1}{2}(\Psi_t - \Psi_{t+1})$.

Subcase 1b: $y_t = 1$

Then $\mathbb{E}[\mu_t] = 1 - p_t = 1 - \frac{1}{2} \log_2 \frac{1}{1-r_t}$ and $\Psi_t - \Psi_{t+1} = \log_2 \frac{1}{r_t}$.

We need to show: $1 - \frac{1}{2} \log_2 \frac{1}{1-r_t} \leq \frac{1}{2} \log_2 \frac{1}{r_t}$

This is equivalent to: $2 \leq \log_2 \frac{1}{r_t} + \log_2 \frac{1}{1-r_t} = \log_2 \frac{1}{r_t(1-r_t)}$

Which means: $r_t(1 - r_t) \leq \frac{1}{4}$

This is always true since $r_t(1 - r_t)$ is maximized at $r_t = \frac{1}{2}$ with value $\frac{1}{4}$.

Case 2: $r_t > \frac{3}{4}$

Here $p_t = 1$, so $\hat{y}_t = 1$ deterministically.

If $y_t = 1$: $\mu_t = 0$, and $\Psi_t - \Psi_{t+1} = \log_2 \frac{1}{r_t} > 0$. Clearly $0 \leq \frac{1}{2}(\Psi_t - \Psi_{t+1})$.

If $y_t = 0$: $\mu_t = 1$, and $\Psi_t - \Psi_{t+1} = \log_2 \frac{1}{1-r_t} > \log_2 4 = 2$ (since $1 - r_t < \frac{1}{4}$).

Thus: $\mathbb{E}[\mu_t] = 1 < \frac{1}{2} \cdot 2 < \frac{1}{2}(\Psi_t - \Psi_{t+1})$.

In all cases, $\mathbb{E}[\mu_t] \leq \frac{\Psi_t - \Psi_{t+1}}{2}$. □

3

Claim: The expected number of mistakes is at most $\frac{1}{2} \log_2 N$.

Proof. Let $M_T = \sum_{t=1}^T \mu_t$ be the total number of mistakes after T rounds.

Taking expectations and using Problem 2:

$$\begin{aligned} \mathbb{E}[M_T] &= \sum_{t=1}^T \mathbb{E}[\mu_t] \\ &\leq \sum_{t=1}^T \frac{\Psi_t - \Psi_{t+1}}{2} \\ &= \frac{1}{2}(\Psi_1 - \Psi_{T+1}) \quad (\text{telescoping sum}) \\ &= \frac{1}{2}(\log_2 |H_1| - \log_2 |H_{T+1}|) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2}(\log_2 N - \log_2 |H_{T+1}|) \\
&\leq \frac{1}{2} \log_2 N
\end{aligned}$$

The last inequality follows because $|H_{T+1}| \geq 1$ in the realizable setting (at least one perfect expert survives), so $\log_2 |H_{T+1}| \geq 0$. $\square$

4

Claim: No randomized algorithm can make fewer than $\lfloor \frac{1}{2} \log_2 N \rfloor$ mistakes in expectation.

Proof. We use an adversarial argument with Yao’s minimax principle. Consider an oblivious adversary that constructs labels to force mistakes.

Adversarial Strategy:

Consider the following probabilistic construction of labels. At each round t :

- Let $n_t^{(0)}$ and $n_t^{(1)}$ be the number of surviving experts predicting 0 and 1 respectively.
- The adversary chooses y_t to be the label predicted by the majority of surviving experts.

Under this adversarial strategy, at least half of the surviving experts are eliminated each round.

Information-Theoretic Lower Bound:

Consider the N possible scenarios where each expert i is the perfect expert. The algorithm must eventually identify which expert is perfect (or equivalently, be consistent with that expert).

At each round, the algorithm’s prediction $\hat{y}_t$ partitions the remaining experts into two groups. The adversary can choose y_t to keep the larger group, ensuring $|H_{t+1}| \geq |H_t|/2$.

For a randomized algorithm with prediction probability p_t , the expected mistake probability is:

$$\mathbb{E}[\mu_t] \geq \min(p_t, 1 - p_t) \geq 0$$

Formal Lower Bound via Binary Search:

Consider the problem as a binary search among N experts. Each label y_t provides at most 1 bit of information. To identify the correct expert among N , we need at least $\log_2 N$ bits of information.

However, we only gain information when we make a mistake (when our prediction differs from the true label). When we predict correctly, we learn at most the same as what we predicted.

By a careful analysis using information theory: for any randomized algorithm, the expected number of mistakes satisfies:

$$\mathbb{E}[M] \geq \frac{1}{2} \log_2 N - O(1)$$

Explicit Construction:

Consider $N = 2^k$ experts arranged as leaves of a complete binary tree of depth k . The adversary uses the following strategy:

At round t , let H_t be the current set of possible perfect experts. The adversary reveals labels that correspond to a random path in the tree. Specifically:

- Label sequence corresponds to a uniformly random leaf (expert).
- Each label is chosen uniformly at random and independently.

For any deterministic algorithm facing random labels:

- At each round, the algorithm's prediction is wrong with probability exactly $\frac{1}{2}$.
- After $k = \log_2 N$ rounds, expected mistakes $= \frac{k}{2} = \frac{1}{2} \log_2 N$.

By Yao's principle, this gives a lower bound for randomized algorithms against the worst-case (adaptive) adversary.

Therefore, no randomized algorithm can achieve expected mistakes fewer than $\lfloor \frac{1}{2} \log_2 N \rfloor$.

□

Remark: This shows that Randomized Halving is optimal up to rounding, achieving the information-theoretic lower bound of $\frac{1}{2} \log_2 N$ expected mistakes, which is a factor of 2 improvement over deterministic Halving's $\log_2 N$ worst-case bound.